Banking Transaction Analytics and Risk Monitoring

Summary

This project aims to analyze banking transactions and understand customer behavior using Machine Learning and Power BI. First, Machine Learning techniques were used to group customers with similar transaction patterns. After creating these customer groups, an interactive Power BI dashboard was developed to visualize transaction activities, customer segments, and possible risk indicators.

The project helps banks understand customer behavior, monitor transaction patterns, and identify customers who may require additional attention due to unusual activities.

Problem Statement

Banks handle a large number of transactions every day. It can be difficult to manually identify unusual transaction patterns or understand the behavior of different customers.

Without proper analysis, banks may miss important information about customer activities, transaction trends, and potential risks. Therefore, there is a need for a system that can automatically group similar customers and present insights in an easy-to-understand dashboard.

This project uses Machine Learning to create customer segments and Power BI to display useful insights through interactive visualizations.

Objectives

- Analyze customer transaction data.
- Group customers with similar transaction behavior using Machine Learning.
- Identify different customer segments based on transaction activities.
- Study transaction patterns, account balances, and login behavior.
- Create an interactive Power BI dashboard for analysis and reporting.
- Support better decision-making through data-driven insights.

Scope

We will analyze the following aspects:

-Transaction-level data: Transaction amount, transaction type, transaction duration, transaction channel, and login attempts.
-Customer-level data: Customer age, customer occupation, account balance, and customer segments generated through Machine Learning.
-Behavioral analysis: Customer transaction patterns, segment-wise transaction activity, and account usage behavior.
-Risk monitoring metrics: Login attempt patterns, high-risk customer groups, and unusual transaction activities.

-Geographic analysis: Transaction locations and location-wise transaction trends

What we're excluding:

-Real-time fraud detection systems.
-Supervised fraud prediction models.
-Credit score and loan analysis.
-External banking or financial datasets.
-Predictive modeling for future transaction behavior.

We assume that the dataset represents historical banking transaction records collected from banking systems and made available through Kaggle for educational and analytical purposes.

Data Sources

-Banking Transaction Dataset for Fraud Detection obtained from Kaggle.

-Customer transaction records containing transaction amount, transaction type, transaction duration, and transaction channel information.

-Customer profile data including age, occupation, account balance, and location details.

-Account activity data containing login attempts and transaction behavior patterns used for risk analysis.

-Customer segments generated through Machine Learning techniques (PCA and K-Means Clustering) based on transaction and customer behavior data.

Methodology

Data Preparation:
-Clean and preprocess banking transaction data.
- Remove unnecessary columns and handle missing or inconsistent values.
- Encode categorical variables such as Transaction Type, Channel, Location, and Customer Occupation.
-Normalize and scale numerical features including Transaction Amount, Account Balance, Transaction Duration, Customer Age, and Login Attempts.
-Apply Principal Component Analysis (PCA) to reduce data complexity and improve clustering performance.
-Use the Elbow Method and Silhouette Score to determine the optimal number of clusters.
-Apply K-Means Clustering to group customers into four distinct segments based on transaction behavior and account activity.

Dashboard Development:
Interactive visuals and filters to include:
-Filters: Customer Segment, Transaction Type, Location, Channel, Customer Occupation.
- Visuals:
- Total Transactions
-Total Transaction Amount
-Average Transaction Amount
-Average Account Balance
-Customer Segment Distribution
- Transaction Type Distribution
-Channel Usage Analysis
-Login Attempts by Customer Segment
-Customer Segment Profile Matrix (Average Transaction Amount, Account Balance, Login Attempts, Transaction Duration, and Customer Age)
- Risk Category Distribution
-Transaction Volume by Location
-Average Transaction Amount by Location
-Age Group Analysis
-Occupation Analysis
-Transaction Amount and Account Balance Comparison Across Customer Segments

Expected Outcomes

-Identification of customer groups with similar transaction behavior.

-Better understanding of transaction trends and customer activities.

-Detection of customer groups with higher risk indicators.

-Easy-to-use dashboard for monitoring transactions and customer behavior.

-Useful insights to support banking decisions and customer management.

-Identification of four distinct customer segments based on transaction behavior.

Tools & Technologies

-Power BI – Interactive dashboard creation and data visualization.
-Power Query & DAX – Data transformation, modeling, and calculation of business metrics.
-Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn) – Data cleaning, preprocessing, clustering analysis, and visualization.
-PCA & K-Means Clustering – Dimensionality reduction and customer segmentation using unsupervised Machine Learning.

-Microsoft Excel – Data validation, exploration, and preparation before analysis.

Risks & Challenges

-Data Preparation: Although the dataset contains no missing values, ensuring that the data is properly cleaned, encoded, and scaled is essential for generating accurate clustering results and meaningful insights.

-Cluster Interpretation: Customer segments generated through Machine Learning may require domain knowledge to accurately understand and label their behavioral characteristics.

-Risk Identification: High login attempts, large transaction amounts, or unusual transaction patterns do not always indicate fraudulent activity and may require further investigation.

-Scalability: Processing large transaction datasets and integrating Machine Learning outputs into Power BI may impact dashboard performance and responsiveness.

Conclusion

The Banking Transaction Analytics and Risk Monitoring Dashboard aims to provide greater visibility into customer behavior and transaction activities within the banking sector. By combining Machine Learning-based customer segmentation with interactive Power BI analytics, it delivers a comprehensive view of transaction patterns, customer characteristics, and potential risk indicators. The insights generated empower financial institutions to make data-driven decisions, improve customer management, strengthen risk monitoring practices, and support safer, more efficient banking operations.